

5

Energy Derivatives: Structures and Applications

5.1 Introduction

Both exchange traded standard derivatives and over-the-counter exotic options are increasingly used in energy markets as a means of controlling exposure to energy prices. In this chapter we give an overview of these options, beginning briefly with some exchange traded derivatives including both futures and options, and continue with a discussion of standard over-the-counter instruments including caps, floors and swap derivatives. We conclude with a discussion of exotic options including compound options, Asian options, lookback options, barrier options, and spread options.

Exotic options are derivatives with a more complicated payoff structure than standard derivatives. They are also known as *second generation*, or sometimes *path-dependent* options. Almost all exotics trade over-the-counter. The distinctive features of the energy markets have led to a range of derivative products which are used for risk management. For example, the high volatility of energy markets and the price spikes observed in electricity markets have made Asian (or average price) options a popular risk management tool. Another popular class of instrument is spread options which are used by many participants to hedge against basis risk. In an article describing the use of exotic options in energy markets Kaminski and Gibner (1995) point out that “one of the main reasons that exotic options have been accepted within the energy industry is that options were, in fact, embedded in many energy contracts long before they became fashionable tools of financial engineering”.

5.2 Exchange Traded Instruments

Although most energy derivatives are traded over-the-counter a number of exchanges worldwide list energy products. For example, the New York Mercantile Exchange (NYMEX) trades futures and American style futures options on oil, gas, and electricity. Chapters 6, 7 and 8 discuss pricing derivatives of this type. NYMEX also lists crack spread options which are options written on inter-market spreads such as the heating oil / crude oil spread-the pricing of these is dealt with in chapters 7 and 8. See also sections 5.3.4 and 5.7.2 for a discussion of crack swaps and crack spread options. Electricity futures and options trade on the Chicago Board Options Exchange with electricity

futures also trading on the Sydney Futures Exchange in Australia and on the Nordic Electricity Exchange.

Although there is an increasing amount of trading in exchange traded derivatives the majority of trading is in over-the-counter instruments. We begin a description of these products by looking at energy swaps.

5.3 Swaps

The first energy swaps were traded in October 1986 when Chase Manhattan Bank acted as counterparty to Cathay Pacific Airways and Koch Industries in an oil-indexed price swap. Other institutions which lead the way in these early days were Citibank, Phibro Energy and Bankers Trust who were to become one of the leading providers of energy derivatives in those early years.

Swaps are also known as Contracts-for-Differences¹ (CFD's) or Fixed-for-Floating contracts. They are used to lock in a fixed price for a certain predetermined but not necessarily constant quantity. In the following sections we describe a wide variety of swap structures which are now common; vanilla, differential, participation, double-up, and extendable.

5.3.1 Vanilla Swap

The vanilla swap is an agreement in which counterparties exchange a floating energy price for a fixed energy price. That is one counterparty pays a fixed price and receives the floating price either by receiving the cash value of the spot energy or the spot energy itself. This counterparty would normally be considered the buyer or receiver of the swap. The other counterparty, the swap provider, receives the fixed prices and either supplies the spot energy or its cash equivalent. The agreement or contract defines the fixed volume or quantity over a specified period of time. The contract is settled at predetermined regular intervals, the settlement or reset dates, over the period of the contract – typically monthly, quarterly, semi-annually or annually. An example of a fixed price receiver might be an oil producer who enters into an oil swap in order to hedge future sales of oil. A producer would be concerned that oil prices could fall in the future and may therefore wish to sell forward its anticipated oil production based on existing oil forward prices. A fixed price payer might be a oil refiner who would be adversely affected by future oil price increases. A swap would allow them to buy forward their anticipated oil consumption based on current oil forward prices.

Vanilla swaps can be priced directly off the forward energy curve since the appropriate portfolio of forward deals is an exact hedge for the swap. Therefore the payoff at each reset date is the same as for a forward contract. Let the reset dates at which the cashflows occur be denoted by s_k ; $k = 1, \dots, m$ and the fixed price payment (strike price) be denoted by K . A vanilla swap is usually viewed as a weighted average of the forwards underlying

¹ Note in some markets caps and floors are also known as Contracts-for-Differences.

the swap with weights being the discount rates to each of the swap settlement dates. Therefore, we write the value of a vanilla swap today (t) as the following²:

$$\text{Vanilla_Swap}(t) = \frac{1}{m} \sum_{k=1}^m P(t, s_k) (F(t, s_k) - K) \quad (5.1)$$

where $P(t, s)$ discounts the future cashflows back to today (see chapter 1).

5.3.2 Variable Volume Swap

This contract is identical to a vanilla swap except that the underlying quantity or volume is not known in advance. For example, in a “Whole of Meter” swap the volume is exactly the metered amount realised for a customer. This type of unknown volume is generically termed “swing” and in order to price contracts involving swing the variability of the volume must be modelled. We discuss variable volume contracts in greater detail in chapter 7.

5.3.3 Differential Swap

A differential swap is similar to a vanilla swap except that the counterparties exchange the difference between two different floating prices (the differential) for a fixed price differential. Differential swaps are often used by companies to hedge basis risk in their normal hedging activities. For example, a refiner’s profitability depends on the market price differential of the raw commodity and the refined products and a differential swap allows a refiner to lock in their refining margin at the fixed price.

A differential swap is essentially a portfolio of forward differential contracts with maturity dates corresponding to the settlement dates underlying the differential swap. As a consequence, the differential swap price is a weighted average of the forward price differences underlying the differential swap, with the weights being the discount rates to each of the differential swap settlement dates. Using the same notation as before but denoting the forward prices on the two different floating prices as $F_1(t, s)$ and $F_2(t, s)$ we have that the time t value of a differential swap is given by

$$\text{Differential_Swap}(t) = \frac{1}{m} \sum_{k=1}^m P(t, s_k) (F_1(t, s_k) - F_2(t, s_k) - K) \quad (5.2)$$

5.3.4 Margin or Crack Swap

The crack spread or margin is the price differential between the raw commodity or energy, for example crude oil, and the refined product or products, for example gasoline,

² Throughout this book we define swap prices on a per reset basis.

jet fuel, or gasoil. The margin or crack swap is a specific form of differential swap in which the fixed price payer, for example a refiner, receives the difference between the market price differential of the raw commodity and the refined products in appropriate fractions and the fixed price. This type of swap therefore allows a refiner to lock in their refining margin at the fixed price. These types of structures are very popular because the market price of the refining margin can often be negative for sustained periods. Crack spreads are discussed further in section 5.7.2 below.

5.3.5 Participation Swap

This is similar to a vanilla swap in that, for example, the fixed price payer is fully protected when prices rise above the agreed fixed price but they participate in a certain percentage of the savings if prices fall. For example, consider a cash settled swap for gasoil at a fixed price of \$150 per tonne with a participation of 50%. If the gasoil price is above \$150 per tonne, say \$160 per tonne, the fixed price payer is compensated by the difference between the market price and the fixed price i.e. receives \$10 per tonne. If the market price is below \$150 per tonne, say \$140 per tonne then the fixed price payer would only pay the provider \$5 per tonne rather than the \$10 per tonne which they would pay under a vanilla swap.

For a fixed payer, this instrument is actually a portfolio of a vanilla swap combined with a long position in a floor (see section 5.4) with a strike equal to the fixed payment and the same settlement dates as the swap. A fixed payer will pay for the floor premium through a higher fixed price on the participation swap than that paid on the corresponding vanilla swap. Participation swaps illustrate that, even in such apparently simple structured swaps, option pricing techniques are important in obtaining correct valuations.

5.3.6 Double-Up Swap

In this type of swap the fixed price payer can achieve a better swap price than the market price but in return the swap provider has the option to double the volume before the pricing period starts. This is achieved by the fixed price payer implicitly selling a put swaption to the swap provider (fixed price receiver). The premium is used to reduce the fixed price of the underlying vanilla swap by splitting it across the swap reset dates and deducting it from the market swap price. A natural fixed price payer would be an oil refiner who would enter into a swap to hedge against future oil price increases. The fixed price payer is exposed to the risk that if swap prices fall the fixed price receiver (swap provider) will exercise the right to double the swap volume. In this case, the fixed payer will now have to pay the relatively high fixed price on twice the notional amount compared to the lower market prices. An oil refiner may be willing to take this risk if they believe oil prices will rise as the swaption will then expire worthless and they achieve a better fixed price on the swap.

5.3.7 Extendable Swap

This agreement is similar to the double-up swap except that the swap provider has the option to extend the period of the swap for a predetermined period. This is again achieved by the fixed price payer selling a put swaption to the swap provider. The premium is again used to reduce the fixed price of the underlying vanilla swap.

5.4 Caps, Floors and Collars

Caps (also sometimes known, together with floors, as one-way CFD's in some markets) provide price protection for the buyer above a predetermined level – the cap price – for a predetermined period of time. A floor guarantees the minimum price that will be paid or received at a predetermined level – the floor price. A collar (also known as two-way CFD's in the Australian electricity market) is a combination of a long position in a cap and a short position in a floor. Standard caps, floors and collars are for predetermined quantities or volumes. These instruments are usually cash settled at regular intervals over the period of the contract.

A generic cap can be viewed as a portfolio of standard European call options with strike prices equal to the cap level and maturity dates equal to the settlement dates of the cap. Let the dates at which the cashflows occur (often called 'fixing' or 'reset' dates) be denoted by t_k ; $k = 1, \dots, m$ and the fixed cap price (strike price) be denoted by K . The time t value of a standard cap is given by

$$\text{Cap_Price}(t) = \sum_{k=1}^m c(t, F(t, t_k); K, t_k, t_k) \quad (5.3)$$

where $c(t, F(t, s); K, T, s)$ is the price of a standard European call option with strike price K and maturity T on a s -maturity forward contract. Results in chapters 6, 7 and 8 show how these option prices can be evaluated for a wide range of energy models. This type of fixed price cap is commonly found in the electricity markets where they are often fixed on an hourly or half-hourly basis. In other energy markets such as oil or natural gas it is more common for the cap to be fixed on a monthly basis and the individual options to be Asian options fixed on the daily price over the relevant month. We describe Asian options in detail in section 5.8.1.

An example of the application of a cap might be an oil refiner who wants to protect themselves against an anticipated rise in the price of crude oil over the next twelve months. Suppose the current price of crude oil is \$18 per barrel and a refiner purchases a monthly cap on the average of the daily spot price in the month struck at \$20 per barrel. A cap with a 1 year maturity on the average daily price over each month is cheaper than a cap on the daily price for a year and achieves roughly the same level of protection due to the mean reversion in oil prices. If the average price in any month exceeds \$20 per barrel then the refiner will still purchase crude oil in the market at the higher price but will be compensated for the difference between the actual price and the strike price by the cap provider.

Similarly, a generic floor is a portfolio of European put options with strike prices

equal to the floor level and maturity dates equal to the settlement dates of the floor. Assuming the same fixing dates as for the cap and with the fixed floor price (strike price) denoted by K , the value of a standard floor at time t is given by

$$\text{Floor_Price}(t) = \sum_{k=1}^m p(t, F(t, t_k); K, t_k, t_k) \quad (5.4)$$

where $p(t, F(t, s); K, T, s)$ is the price of a standard European put option with strike price K and maturity T on a s -maturity forward contract.

A collar is simply a portfolio of a long position in a cap and a short position in a floor. Collars are typically used by energy buyers who wish to hedge against price increases and wish to use the premium on short floors to pay for the cap protection. For example, a buyer who decides they cannot afford to pay the full price of the cap might agree to purchase a cap and finance it through the sale of a floor. The strike prices of the cap and floor can be set to yield a collar at zero cost.

5.5 Swaptions

A swaption is a European option on an energy swap. A call option on the swap, also called a payer swaption, with strike price K and time to maturity T , provides the holder of the option the right to enter into a swap, paying the fixed price K and receiving the floating energy spot price. Conversely, a put option on a swap, or receiver swaption, gives the purchaser of the option the right to sell a swap or receive the fixed price K and pay the floating price. If $s_k; k = 1, \dots, m$ denote the dates at which the swap cashflows occur, we can write the payoff to the payer swaption as

$$\max \left(0, \frac{1}{m} \sum_{k=1}^m (F(T, s_k) - K) \right) \quad (5.5)$$

Note that equation (5.5) represents the payoff to a cash settled option. If, upon exercise of the option, the option holder enters into a new ‘physical’ swap position, the payments will be exchanged on the swap rest dates and hence the payoff at time T must reflect these future cashflows:

$$\max \left(0, \frac{1}{m} \sum_{k=1}^m P(T, s_k) (F(T, s_k) - K) \right) \quad (5.6)$$

where $P(T, s_k)$ is the value of a pure discount bond maturing at time s_k , observed at the maturity of the option, and corresponds to the initial discount function unless we assume stochastic interest rates which are correlated with the energy price. The receiver swaption can be written similarly but as a put option for both cash settled and physical settled options.

The value of a cash settled European payer swaption can therefore be written in terms of the discounted expected payoff at maturity;

$$Swpn(t; K, T, \{s_k\}, m) = P(t, T)E_t \left[\max \left(0, \frac{1}{m} \sum_{k=1}^m (F(T, s_k) - K) \right) \right] \quad (5.7)$$

If the payer swaption involves physical settlement then equation (5.7) becomes:

$$Swpn(t; K, T, \{s_k\}, m) = P(t, T)E_t \left[\max \left(0, \frac{1}{m} \sum_{k=1}^m P(T, s_k)(F(T, s_k) - K) \right) \right] \quad (5.8)$$

Since a swap is a portfolio of forward deals, a swaption is an option on a portfolio of forwards, therefore analytical formulae are typically not available for these instruments and numerical techniques must be used. Sections 7.2 and 8.8 provide a discussion on how swaptions can be priced, in a single factor spot price model and a general multi-factor forward curve model, respectively.

Swaptions are typically purchased by institutions that anticipate the need for fixed price protection of a swap but are unsure of the direction of future market prices. If the institution purchased a forward starting swap then it would be locked into current swap prices and if the market prices fell would not be able to participate in the savings. By purchasing a swaption the institution can lock into the current market swap prices by exercising the option when markets move favourably. However, if market prices turn out to be lower then the institution can abandon the option and obtain the required swap more cheaply directly from the market.

5.6 Compound Options – Captions and Floptions

An option which allows its holder to purchase or sell another option for a fixed price is called a compound option. Compound options occur in energy markets in the form of captions and floptions as well as in their standard form as call or puts on simple calls or puts. A caption is an option on a cap and a floption is an option on a floor. These instruments are therefore options on portfolios of options and as such no simple analytical formulae exist³.

Compound options are typically purchased by companies which are anticipating the need for option like protection e.g. a cap or floor in the medium term future and are concerned that market prices may rise between now and the time when they will need to purchase the protection. A typical example is the case of a company which is bidding for a project at a certain price. They are exposed to the risk that market prices will be higher than those on which the bid was based. The company would therefore like to purchase an option which locks in the prices on which the bid was based but only if they are awarded the contract. By purchasing an option on the option, struck at the relevant forward price, they can pay a small premium now to lock into a known price for the future protection.

³ European versions of standard calls or puts on underlying calls or puts can be priced using the results of Geske (1979) in a Black-Scholes world.

5.7 Spread and Exchange Options

A spread option is written on the difference between two prices. Options of this type are popular in the energy industry as many players are exposed to the difference in prices of related energies or are able to sell the energy in different markets. Exchange options are also designed to provide a payout which is based on the relative movement of two energies.

We begin by considering two types of spread options. The first is an option written on the spread of two different maturity futures contracts on the same underlying spot instrument. The second is an option on the spread between the prices of futures contracts of two related commodities.

5.7.1 Calendar Spreads

If the futures contracts are written on the same underlying energy, but with different maturity dates, then the option is often referred to as a *calendar spread* option. Let $F(t, s_1)$ and $F(t, s_2)$ represent the current price of futures contracts on an energy (say oil) that mature at times s_1 and s_2 respectively. We define the maturity payoff to a European call spread option with strike K and maturity T ($t \leq T \leq s_1 \leq s_2$) to be;

$$\max(0, F(T, s_1) - F(T, s_2) - K) \quad (5.9)$$

The payoff to a European put spread option is defined similarly. Such options are used to protect against changes in the shape of the energy forward curve. The value of the option at time t of the call option can be written generally as the following discounted expectation;

$$\text{Calendar_Spread}(t; K, T, s_1, s_2) = P(t, T)E_t[\max(0, F(T, s_1) - F(T, s_2) - K)] \quad (5.10)$$

These options are valued in chapters 7 and 8 under single factor and multi-factor model assumptions respectively.

5.7.2 Crack Spreads

If the futures contracts underlying the option are written on two separate energies (say natural gas and electricity), then the option is often referred to as a *crack spread* option. The maturity of the futures contracts can be on the same date or different dates. Options of this type are often used by companies who are exposed to the difference in price between two different energies. For example, one energy might be an input into a process that produces another energy, as is the case for gas fired electricity generators. Other examples include heat spreads (the difference between Number 2 heating oil and crude) and gasoline spreads (the difference between unleaded gasoline and crude).

Let $F_a(t, s_1)$ represent the price of an s_1 -maturity futures contract on energy a (say natural gas) and $F_b(t, s_2)$ represent the price of an s_2 -maturity futures contract on energy b (say electricity). The payoff to a European call option with maturity T and strike K on

the spread between these two forward contracts can be represented by the following;

$$\max(0, F_a(T, s_1) - F_b(T, s_2) - K) \quad (5.11)$$

and so the value of the option at time t can be written generally as;

$$\text{Crack_Spread}(t; K, T, F_a, F_b, s_1, s_2) = P(t, T)E_t[\max(0, F_a(T, s_1) - F_b(T, s_2) - K)] \quad (5.12)$$

We show in chapter 8 how options of this type can be valued using a multi-energy forward curve model.

5.7.3 Exchange Options

The spread options of this section have a payoff which is determined by the difference between two forward contracts. Exchange options, on the other hand, provide a payout which is based on the relative performance of two energy prices. We identify two types of exchange option. The first, often called an ‘out-performance’ option pays off on the better performing energy. For example, for a T -maturing European call out-performance option on the forward contracts in the previous section, the maturity value can be determined as;

$$\max\left(0, \left(\frac{F_a(T, s)}{F_a(t, s)} - 1\right), \left(\frac{F_b(T, s)}{F_b(t, s)} - 1\right)\right) \quad (5.13)$$

The out-performance option pays out if one energy price moves more than the other in a given direction, for example if gas prices rise more than oil prices.

A second exchange option has a similar payoff to a crack spread option (c.f. equation (5.11)), but again on the percentage movements in the energy prices;

$$\max\left(0, \frac{F_a(T, s)}{F_a(t, s)} - \frac{F_b(T, s)}{F_b(t, s)} - K\right) \quad (5.14)$$

Both of these options are useful in situations where it is the relative movement of two energy prices which is important.

5.8 Path Dependent Options

Although many of the options that we have described so far have complicated maturity payoffs, they have been independent of the path taken by the spot or futures prices over the life of the option. The options that we describe in this section are often known as path dependent options because the payoff to this type of option depends on the path that the underlying energy follows before maturity (or some part of the options life) and not solely on the terminal price⁴. For the majority of traded path dependent options, if not

⁴ Path dependent options are often treated as synonymous with exotic options but are really a subclass.

all, the fixings for the dependency occur at discrete times. In the following examples we assume a consistent set of predetermined dates t_k ; $k = 1, \dots, m$, on which prices are observed. S_k will denote the spot price at time t_k . If the path dependency is observed over the whole life of the option then $t_1 = t$ and $t_m = T$, but in general it can be a shorter period. Usually the fixing dates will be equally spaced such as hourly, daily, weekly or monthly but they can be a predefined set of ad hoc dates. It is also not uncommon for the spot price fixings to be replaced by observations of particular forward or futures prices. The most well known examples of path dependent options are Asian (or average price) options, barrier options, and lookback options.

5.8.1 Asian Options – Average Price and Average Strike

Asian options are options whose final payoff is based in some way on the average level of an energy price (spot, forward or future) during some or all of the life of the option. Of the range of exotic options, Asian options in the equity and commodity markets are priced and risk managed as being almost vanilla in that they are now one of the best understood options and are very popular as a means to hedge exposure⁵. There are two basic styles of Asian option; average price options (sometimes called average rate options due to their origin in the foreign exchange markets) and average strike options⁶. Several forms of averaging are possible with typically the arithmetic average being specified, a convention that we follow in this section. The payoffs to the four main standard Asian options are summarised in table 5.1, where K is the strike price for the fixed strike options.

TABLE 5.1 Payoffs of the Standard Asian Options

Name	Payoff
Average Price Call Option:	$\max\left(0, \frac{1}{m} \sum_{k=1}^m S_k - K\right)$
Average Price Put Option:	$\max\left(0, K - \frac{1}{m} \sum_{k=1}^m S_k\right)$
Average Strike Call Option:	$\max\left(0, S_T - \frac{1}{m} \sum_{k=1}^m S_k\right)$
Average Strike Put Option:	$\max\left(0, \frac{1}{m} \sum_{k=1}^m S_k - S_T\right)$

⁵ See Levy (1997) for the uses, risk management, and numerical techniques for pricing Asian options in a Black-Scholes world.

⁶ Average price options are also known as ‘fixed strike Asians’ whilst average strike options are also known as ‘floating strike Asians’.

For existing Asian options, the averaging period may have started before the current valuation date – in this case the ‘current’ average at the time of contract valuation is needed for pricing. Asian options can be both of European or American style exercise. We give an example of how to price both in sections 7.2 and 7.5 respectively.

As we have already seen one of the commonest occurrences of Asian options is as the component options in caps and floors. In general the main use of Asian options is hedging an exposure to the average price over a period of time. For example, large buyers of electricity often require to hedge their average fuel cost as the prices they charge customers are based on the average purchase prices. Asian options also fit the risk profile of energy producers who need to meet budget targets based on average prices.

5.8.2 Barrier Options

In recent years one of the most significant growth areas in new option types has been in the area of barrier options. Barrier options are standard options that either cease to exist (knock-out) or only come into existence (knock-in) if the underlying price (usually a spot energy price or the price of a forward contract) crosses a predetermined level – the barrier (which we denote by H). Barrier options were originally introduced because they are cheaper than otherwise identical standard options due to the fact that the option can cease to exist or never come into existence. There are four basic types of barrier options – the barrier can be above the current underlying price (up barriers) or below (down barriers) and the option can cease to exist (knock-out) or come into existence (knock-in) when the barrier level is crossed. Each of these four types can be either a call or a put which leads to eight different barrier options. A standard variation of the options just described are options which pay a predetermined cash rebate if an ‘out’ option disappears or an ‘in’ option never appears.

As a more detailed illustration we describe a down and out call option. Figure 5.1 shows two possible paths for the spot energy price. Both begin at the same spot price and end at the same spot price which is above the assumed strike price of the option. However, the lower price path crosses the barrier level (H) from above and so in this case would ‘knock-out’ and be worthless.

Analytical formulae exist for the basic barrier options in the Black-Scholes-Merton world described in chapter 1 under the assumption that the crossing of the barrier is continuously checked⁷. For discretely fixed barrier options we need to use numerical techniques to evaluate prices. The common names and payoffs of the eight basic barrier options are summarised in table 5.2.

⁷ See Rich (1994) for the mathematical foundations of barrier option pricing. However, his exposition is centred on a generalised BSM world.

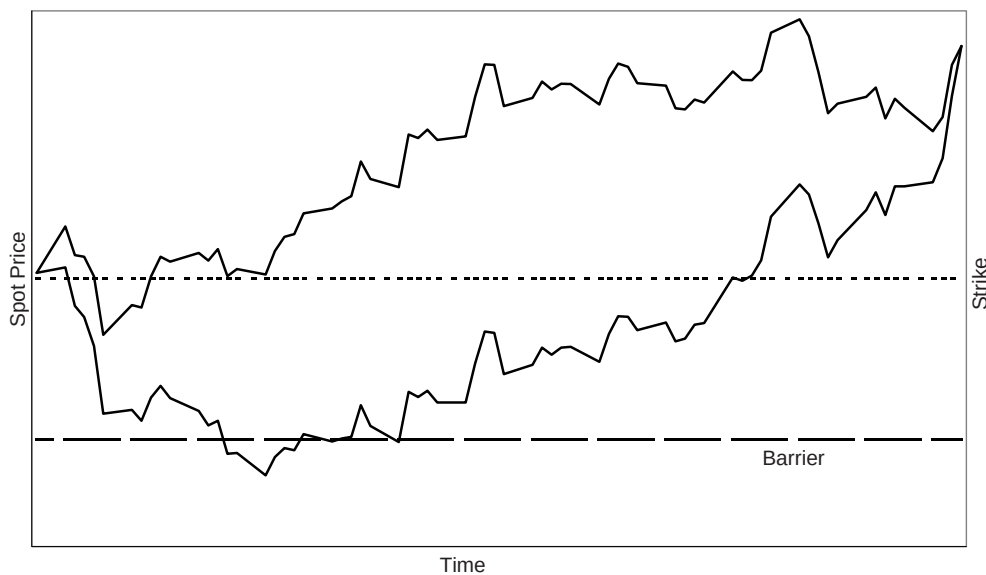


FIGURE 5.1 Down and Out Call Option Example

TABLE 5.2 Payoffs of Standard Barrier Options

Name	Payoff
Down and Out Call	$\max(0, S_T - K) 1_{\min(S_1, \dots, S_m) > H}$
Up and Out Call	$\max(0, S_T - K) 1_{\max(S_1, \dots, S_m) < H}$
Down and In Call	$\max(0, S_T - K) 1_{\min(S_1, \dots, S_m) \leq H}$
Up and In Call	$\max(0, S_T - K) 1_{\max(S_1, \dots, S_m) \geq H}$
Down and Out Put	$\max(0, K - S_T) 1_{\min(S_1, \dots, S_m) > H}$
Up and Out Put	$\max(0, K - S_T) 1_{\max(S_1, \dots, S_m) < H}$
Down and In Put	$\max(0, K - S_T) 1_{\min(S_1, \dots, S_m) \leq H}$
Up and In Put	$\max(0, K - S_T) 1_{\max(S_1, \dots, S_m) \geq H}$

Where, the indicator function is defined as follows

$$1_{\min(S_1, \dots, S_m) > H} = \begin{cases} 1 & \min(S_1, \dots, S_m) > H \\ 0 & \text{otherwise} \end{cases} \quad (5.15)$$

The other indicator functions are defined similarly. Barrier options can be either European or American style exercise and we show how to price both types in chapters 7 and 8.

Barrier options are used to obtain cheaper hedges than are provided by standard options. For example, consider an energy producer with a natural long position in the underlying energy. They would potentially be looking to purchase an out-of-the-money put option to hedge the risk of the market price of the energy falling. However, if the market price of energy rises, a standard put option hedge would become worthless as the strike price of the option would be far too low. If instead the energy producer entered into an up-and-out put, it would be less expensive than the standard put whilst still providing the same protection. Furthermore, when energy prices rose sufficiently, the up-and-out put would extinguish automatically and could then be replaced by another with a more appropriate strike price.

5.8.3 Lookback Options – Fixed Strike and Floating Strike

With lookback options the payout is a function of the highest or lowest price at which the underlying asset trades over some period during the life of the option. For this reason they are occasionally referred to as hindsight options. There are two basic styles of lookback option; fixed strike and floating strike options⁸. Table 5.3 summarises the payoffs of the standard lookback options.

As with the other path dependent options of this section, the spot fixings are often replaced by observations of particular forward or futures prices. Notice that floating strike lookback options are not really ‘options’ at all as the worst that can happen is that the index value finishes at the minimum point (for a call) or the maximum point (for a put).

Figure 5.2 illustrates a possible price path and the payoff to a floating strike lookback call option. The payoff is equal to the difference between the final spot price and the minimum spot price over the life of the option.

Analytical formulae exist for the standard lookback options in a Black-Scholes-Merton world when the maximum or minimum is checked continuously, but numerical techniques must be used for discretely fixed examples. Lookback options can be of European or American style exercise. European style lookbacks for energy models are

TABLE 5.3 Payoffs of Standard Lookback Options

Option Name	Option Payoff
Fixed Strike Lookback Call	$\max(0, \max(S_1, \dots, S_m) - K)$
Fixed Strike Lookback Put	$\max(0, K - \min(S_1, \dots, S_m))$
Floating Strike Lookback Call	$\max(0, S_T - \min(S_1, \dots, S_m))$
Floating Strike Lookback Put	$\max(0, \max(S_1, \dots, S_m) - S_T)$

⁸ See Heynan and Kat (1997) for the uses, risk management, and numerical techniques for pricing lookback options in a BSM world.

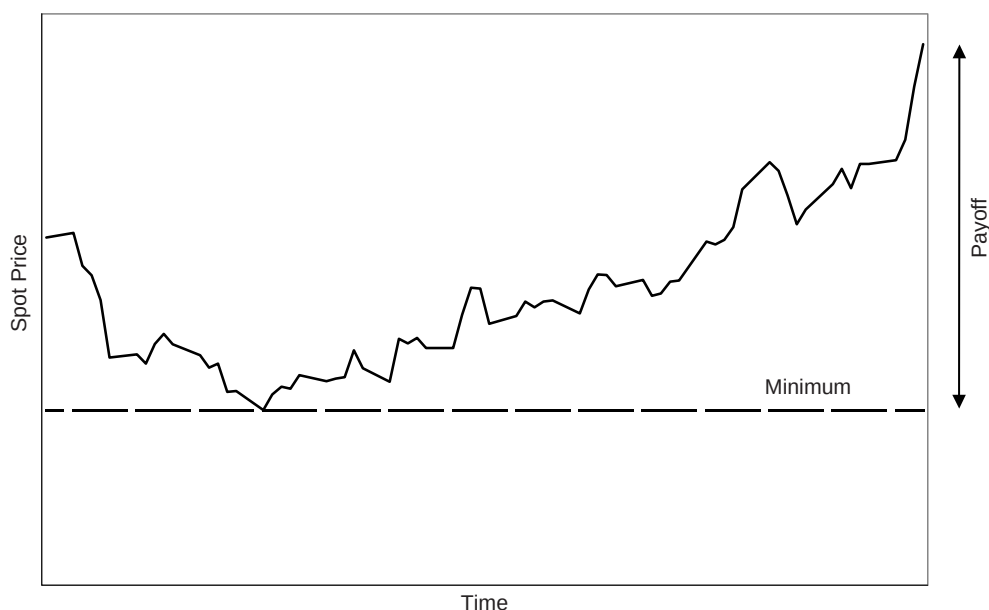


FIGURE 5.2 Example Payoff to a Floating Strike Lookback Call Option

priced in section 7.2 via Monte Carlo simulation and American style options can be priced using the techniques of section 7.5.

Lookback options tend to be less popular than the other exotics we have described so far because they allow the holder to effectively buy at the minimum or sell at the maximum price and so are relatively expensive. However, floating strike lookbacks do occur in a form in which the period over which the maximum or minimum is observed is a fairly short period e.g. one month. A type of lookback option has also appeared recently in electricity markets which exhibit sudden and relatively large price spikes. These are lookback caps which cap only a relatively small number of the highest prices in a given period. For example a lookback cap might have the same structure as the electricity cap described in section 5.4 but instead of capping every half hour electricity pool price over a certain cap level the product caps only the five highest prices in a given period.

5.8.4 Ladder and Cliquet Options

Ladder and cliquet options are discrete level and time versions of lookback options respectively. Depending on the level of discreteness they can be much cheaper than lookback options and are therefore relatively more popular.

Ladder options have a predefined set of levels L_k ; $k = 1, \dots, n$ such that if the underlying price crosses a particular level it locks in a minimum payoff equal to the

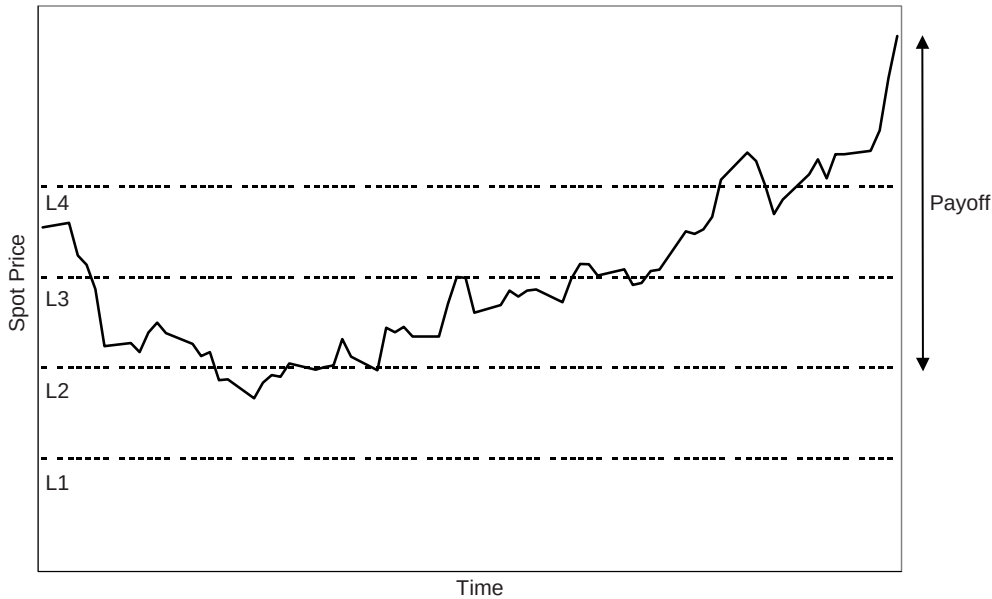


FIGURE 5.3 Example Payoff to a Floating Strike Ladder Call Option

difference between the level crossed and the strike price. Similarly to lookback options there exist fixed strike and floating strike ladder options for both calls and puts. Figure 5.3 illustrates a possible price path and the payoff to a floating strike ladder call option. Unlike the analogous lookback option, the floating strike ladder option ‘locks in’ the lowest level L crossed and not the absolute minimum. The maturity payoffs to fixed and floating strike ladder calls and puts are shown in Table 5.4.

Ladder options can be replicated with a portfolio containing the standard European option with the same strike price and for each ladder level a pair of up-and-in options – for example, for the ladder call; a short put with a strike equal to the ladder level (or strike) below and a long put with a strike equal to the ladder level.

Cliquet options have a predefined set of dates at which the underlying price is observed

TABLE 5.4 Payoffs of Ladder Options

Option Name	Option Payoff
Fixed Strike Ladder Call	$\max(0, S_T - K, (L_1 - K)1_{\max(S_1, \dots, S_m) > L_1}, \dots, (L_n - K)1_{\max(S_1, \dots, S_m) > L_n})$
Fixed Strike Ladder Put	$\max(0, K - S_T, (K - L_1)1_{\min(S_1, \dots, S_m) < L_1}, \dots, (K - L_n)1_{\min(S_1, \dots, S_m) < L_n})$
Floating Strike Ladder Call	$\max(0, (S_T - L_1)1_{\min(S_1, \dots, S_m) \leq L_1}, \dots, (S_T - L_n)1_{\min(S_1, \dots, S_m) \leq L_n})$
Floating Strike Ladder Put	$\max(0, (L_1 - S_T)1_{\max(S_1, \dots, S_m) \geq L_1}, \dots, (L_n - S_T)1_{\max(S_1, \dots, S_m) \geq L_n})$

and they payoff the maximum of the differences between these fixings and the predefined strike price for a call or the maximum of the differences between the predefined strike price and the fixings for a put. Cliquet options are therefore identical to the fixed strike lookbacks defined in section 5.8.3.

5.9 Summary

In this chapter we have discussed some of the structures used in the energy derivatives markets and their applications. Compared with many other markets, exotic options are almost common place in the energy markets. The high volatility exhibited by many energy commodities, especially the price spikes observed in the power markets, and the fact that they have been embedded in many energy contracts for a long time has led to a ready acceptance for these kinds of derivatives. Because of the diverse nature of the products used we have only tried to give examples of the different categories of exotics and have not attempted to provide an exhaustive list.